Autonomous Buildings: Smart Savings

Six steps to operations and maintenance efficiencies through autonomous smart buildings management.

SIEMENS

Contents

Automation and Autonomy

Across the world, there is a widely recognized need for commercial buildings to save energy, reduce costs and reduce carbon emissions. The great enabler for these objectives is to convert buildings to be digitalized, smart and – the ultimate goal - 'autonomous'. Autonomous buildings can self-manage, responding to issues and occupant needs, and optimizing their performance, without human intervention.

> A smart building is much more than its IT hardware, software and sensors. Its intelligence comes from the connectivity of its systems and the data collected on a building's energy use, temperature, lighting and other everyday operations.
>
> Turner & Townsend

There are different levels of autonomy from making recommendations to automatically taking actions. This allows agile management of energy consumption, heating and lighting, access management and safety control. The result is both a more attractive environment for occupiers, along with considerable reduction in running costs. Not surprisingly, the concept has been applied across single buildings, to campuses, to global portfolios, to whole cities, with governmental policy and global associations dedicated to the autonomous transition.

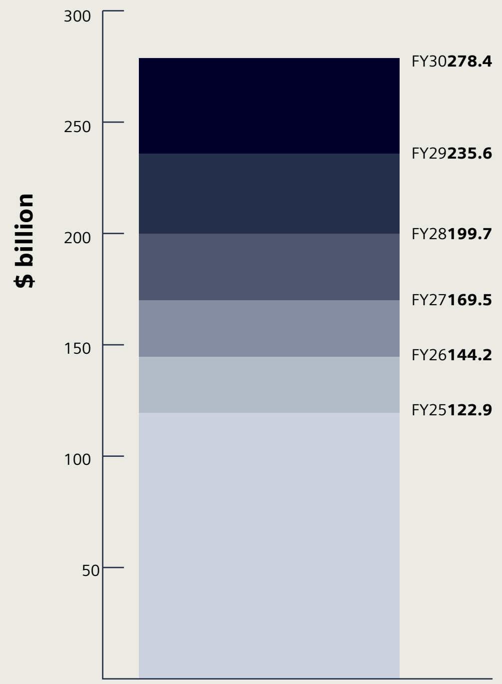
Construction rules and regulations in Asia, Europe and the USA alike demand that new buildings be made sustainable and energy-efficient, something that is only possible using smart technology, data and digitalization strategies.

Yet by far the greatest energy consumption and carbon emissions come from the existing building stock. In Europe, around 80% of commercial buildings were constructed before the year 2000[1]: in the US the proportion is approximately 75%[2].

Given that there are now mandatory corporate energy-efficiency reporting rules in Europe and China, companies want to occupy buildings with a lower carbon footprint, thereby reducing their own overall emissions. Moreover, across the world, energy-efficiency cost savings make buildings more attractive. For the commercial property industry, the healthcare sector, higher education and public estate alike, that puts on the pressure to manage energy consumption better through smart, unified, digital technology.

It is hardly surprising, then, that the pace of autonomous building development is roaring ahead and is unlikely to slow down. Our chart[3] shows the expected pace of growth.

Global Smart building Tech Market 2025-2030[3]



The commercial drivers of autonomous smart buildings

Legislation, regulation and compliance are mandatory drivers of digitalization in the built environment. However, if commercial benefits are also gained, the business case becomes doubly compelling. Strong business cases are built on hard commercial advantage. Indeed, commercial benefit can drive smart building adoption irrespective of any sustainability or regulatory drivers. In the case of autonomous smart buildings, those commercial benefits are twofold: cost-efficiency and profitability.

Cost-efficiency. Most importantly, the cost of managing and maintaining buildings can be radically reduced with smart, coordinated, unified controls, offering visibility, insights and ability to take action:

- Real-time, automated, fault detection
- Remote triage and path to resolution
- AI-driven capabilities to prevent breakdowns before they happen
- Condition-based maintenance, rather than scheduled routines
- A scalable, holistic view to manage multiple portfolio buildings
- Embedded knowledge databases to support staff resolving issues

Competitiveness. By becoming smart, autonomous buildings also become more valuable to their owners. In healthcare, autonomous buildings enable greater workflow and operational cost efficiency. The same is true on university campuses.

And in commercial property, landlords, owners and portfolio managers are in a marketplace today where changing occupancy patterns (such as hybrid work) have led to over-supply. So those commercial buildings need to compete harder to attract tenants and maintain (or even increase) their value. Central, remote, virtual control of buildings improves:

- Occupier comfort and satisfaction
- Building equipment (e.g. heating ventilation & air-conditioning, pumps, air handling unit) uptime and resilience
- Sustainability goal achievement
- Flexibility to manage variable occupancy patterns
- Safety, access and security responsiveness

BIoT [Building Internet of Things] platforms have evolved from optional enhancements to core infrastructure components for modern real estate. They deliver significant operational value by integrating previously siloes systems, aggregating and contextualizing data across multiple properties, enabling centralized visibility and control, and creating a flexible foundation for future technology adoption without requiring wholesale system replacements. ***Memoori, IoT Platforms in Smart Commercial Buildings***[4]

Even in hybrid work environments, occupancy fluctuates, not just by day but by where people are in the building – vertically and horizontally. So, smart technologies are helping to optimize that. Think of your smart building data as jet fuel. ***CBRE***[5]

Companies are opting for high-quality premises in prime locations to meet higher energy-performance standards and to provide an attractive work environment for employees.

Consequently, well-located, modern buildings that comply with environmental, social, and governance (ESG) standards are expected to outperform other assets ***Oxford Economics***[6]

In the rapidly evolving healthcare landscape, the integration of Internet of Things (IoT) and Artificial Intelligence (AI) is transforming hospital efficiency... IoT devices enable real-time monitoring of patient health and hospital assets, facilitating timely interventions and maintenance. ***Journal of Computer Science Application and Engineering***[7]

Spaces within university estates will become versatile, equipped with smart technologies... Connected technology will also allow for estates to be operated more efficiently, informed by constant data collection. ***PwC***[8]

Facing the **Challenges**

Clearly, asset owners want platforms that allow buildings operations and maintenance (O&M) to be cost-efficient, and which make the buildings portfolio more competitive and valuable. However, solutions have to combat five clear challenges.

Challenge #1 – Inflation

The largest proportion of costs in a building portfolio comes from the operational stage – up to 80% when maintenance and utilities are taken into account[9]. The steep inflationary rises in O&M costs since the beginning of the decade have been well documented[10]. And they are expected to continue rising, with one country estimating a rise just short of 20% to the end of the decade[11]. Containing costs is therefore a major priority, especially unplanned downtime. According to the US Environmental Protection Agency, every dollar spent in the education sector on preventive maintenance yields $4 in savings by avoiding the costs of future repair or replacement of building systems[12].

Challenge #2 – Buildings Technology Silos

Most buildings today operate with a patchwork of systems. Even worse, sub-metering, unit controls and peripheral systems, which are critical for granular performance insight are often offline or unconnected. A UK survey found that 70% of asset owners struggle with building safety and maintenance due to disparate systems and formats, hindering effective golden thread management[13].

Challenge #3 – Changing Occupancy Patterns

Hybrid and remote working appears to be persisting across the world[14]. This is not just in business, but also higher education[15], healthcare[16] and the public sector. Indeed, A recent study found that 42% of European job seekers would reject a job without remote options[17]. A shift to remote working is likely to wipe off $800 billion from the value of office buildings in major global cities by 2030, according to a study published by consulting firm McKinsey[18]. In Asia, this is fueling greater demand for ‘premium' business space[19], which by definition means smarter business buildings.

Challenge #4 –Skills Shortages

JLL's 2024 Future of Work Survey found that 75% of facilities managers report difficulty hiring and retaining skilled technicians, especially those with cross-disciplinary IT/OT and energy management knowledge[20]. In a CBRE survey, there was a notable increase in the number of clients requesting data literacy as a core skill of facilities managers working on their sites[21]. The U.S. Bureau of Labor Statistics projects that building and general maintenance roles will grow 5% between 2022 and 2032, yet demand is already outpacing supply in high-skill areas like automation and predictive maintenance[22].

Challenge #5 – Meeting Sustainability Targets

The EU's Energy Performance of Buildings Directive mandates renovation of the worst-performing 15% of non-residential buildings by 2030, a further 10% by 2034, with a zero-emissions sector by 2050[23]. Some countries are ahead of the Union[24], with mandatory energy performance (EPC) already making the lowest classes of building un-leasable without renovation. In The US and Asia, state-level regulations are also demanding minimum performance standards. Moreover, corporate sustainability reporting standards in the US, Europe and China are driving demand for highly sustainable buildings[25].

The dream	The nightmare
Autonomous smart buildings management	**No autonomous smart buildings transformation**
Greater productivity from operations and maintenance staff through virtual tools, automation and insights, allowing them to do more with less and supporting skills and knowledge gaps.	Escalating operations and maintenance costs and inefficiency as estate/portfolio expands with difficulty finding scarce skilled staff
Scalable, smart building control that grows with the organization without escalating costs	Growing costs and inefficiencies as an estates footprint expands along with increasing lack of control and insight
Remote cloud monitoring and control from anywhere for efficient, fast problem resolution and remote environment adjustment	Manual inspections and expensive reactive repair once equipment fails, along with unplanned downtime and disrupted occupant experience
Real-time – often automated - adjustment of energy consumption to satisfy needs while minimizing cost	Standard energy settings, not tailored to changing occupier requirements, leading to unnecessary consumption & waste
Agile, autonomous building control to enhance tailored occupier experience while minimizing cost – increases asset attractiveness and profitability	Loss of competitive value of assets as occupiers seek premium commercial properties that can accommodate variable occupancy patterns and hybrid working
Automated sustainability and other regulatory reporting that scales affordably with growing compliance demands	Manual sustainability and regulatory reporting where costs keep rising with increasing compliance requirements

In one example feature of autonomous smart buildings management, **Siemens Building X** has released **fault detection and diagnosis (FDD)** module.

It facilitates early digital detection of faults through the continuous application of rules on equipment parameters.

It also enables the user to analyze and triage faults remotely.

This enables an organization to move from scheduled, regular maintenance to a condition-based approach – triggered by early-warning, pre-failure warning signs.

The result is more uptime (better occupier experience), reduced maintenance costs and extended equipment lifespan. Acting on FDD-driven findings can also reduce energy consumption by 5-20%[26].

What size the prize?

We talk about cost efficiency from smart buildings management. But what does that really mean? Let's get a sense of the scale of financial savings to be made? After all, this is the starting point for commercial real estate professionals and organization CxOs alike.

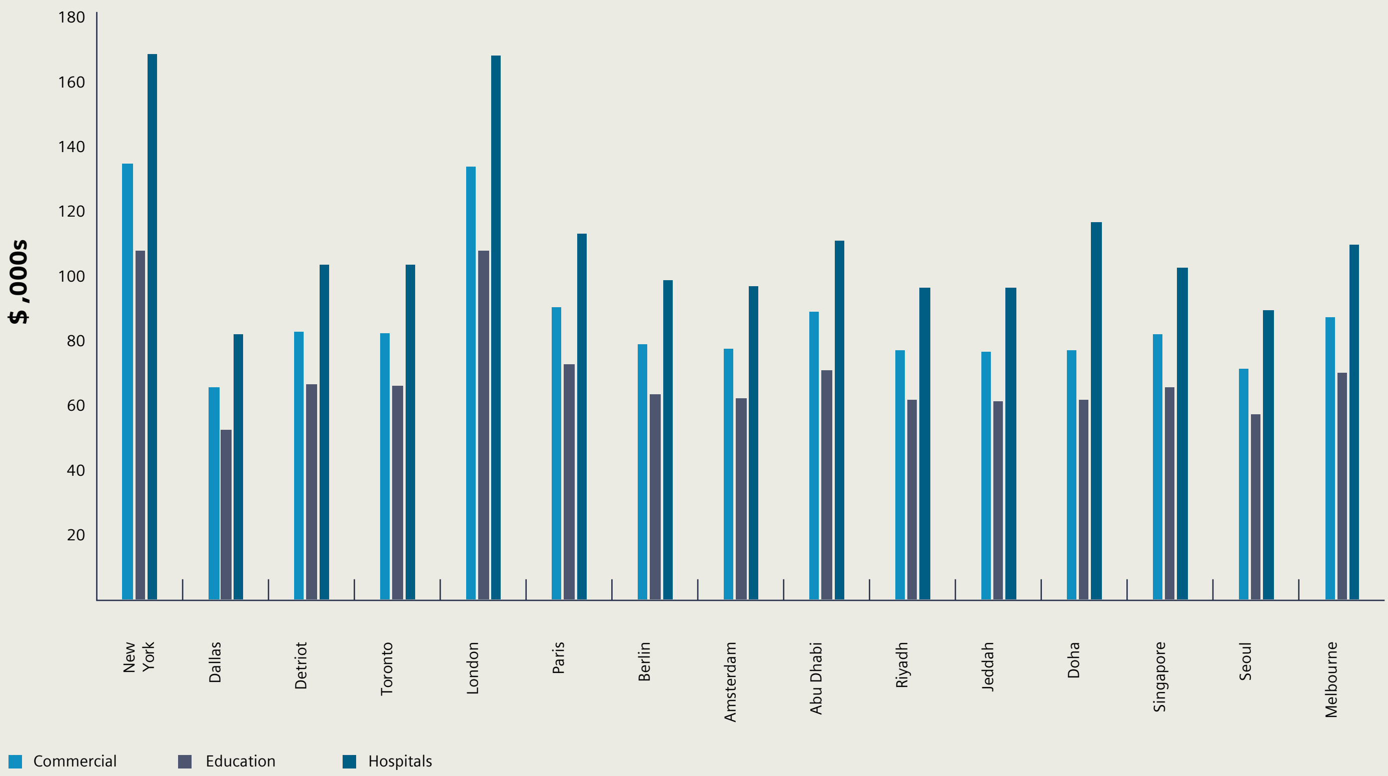
For organizations considering the transition to smart autonomous building management, this paper presents a conservative estimate of potential O&M savings using a reference scenario of a typical 10,000 m² building[27].

The illustration is designed to provide readers with a clear sense of the potential scale of impact across their own buildings or portfolios.

The estimate is based on third-party benchmarks for typical annual maintenance costs and assumes an average O&M savings of approximately 20% through the adoption of autonomous smart building management solutions[28].

While every building and portfolio has to be assessed on its individual merits, the chart above provides a useful scoping point and geographical comparison for asset owners and managers.

Typical annual savings from autonomous smart building management for a 10,000m2 building[27]



Next steps - What to demand from Autonomous Systems

We have established that the drive to smarter buildings has strong commercial paybacks.

It saves O&M costs, improves the attractiveness of buildings for tenants and occupiers, enhances the value of assets and asset portfolios, and offers scalability for portfolio growth and regulatory compliance.

So the next step is to decide which digital building platform to choose. Which solution will best bring all your buildings systems together into a single centralized view and control center – accessible by all stakeholders wherever they are.

As asset owners and managers evaluate solutions for the virtual management of their buildings, there are six major criteria that must be in place for a successful transition to smart O&M.

As a handy reference document for professionals assessing their journey to smart buildings management, we have summarized them below.

The architecture of BIoT (Building IoT) platforms follows a structure with four primary layers. The Source Data Layer collects and harmonizes input from diverse building systems... The Edge & Connectivity Layer manages device communications across multiple protocols...

The Data Processing & Intelligence Layer handles data storage... and applies rules-based automation and emerging machine learning... The Application & Interface Layer delivers user-facing tools including dashboards, mobile apps and, increasingly, low code environments...[29]

Memoori, IoT Platforms in Smart Commercial Buildings

Evaluating autonomous smart solutions: Six steps to success

1. Is your physical environment **sufficiently digitized?**

The foundations for centralized, virtual management of buildings start with digital connectivity for physical assets and analog processes such as sensor data, asset location tracking, asset logs, maintenance records, access permissions, safety systems, and so on. A key first step is to audit the current state of building asset and systems digitization to understand the initial foundational investment required. Any digitization shortfall at this stage will amplify into major issues further down the line.

Those without digital twins of their core assets or operations will face questions about their control, governance and reporting.

KPMG, The Great Reset: Emerging trends in infrastructure and transport[30]

2. Are those systems **interconnected and unified?**

All BMS (building management systems), IT, and OT data needs to flow into a single, unified platform, eliminating the inefficient world of fragmented tools and siloed teams. Major players in the industry now offer single-version-of-the-truth, comprehensive cloud-based suites designed to break down the barriers between building controls and business applications. BIoT is being integrated with Integrated Workplace Management Systems (IWMS), Computer-aided Facilities Management Systems (CFM) and Computerized Maintenance Management Systems (CMMS).

There is a decisive shift in the sector towards open architecture – an API-first approach with wide interoperability support that embraces legacy protocols and newer IoT data formats.

The outcome is that every stakeholder can view, monitor, and optimize building performance from any location.

Siemens Building X seamlessly integrates with new and existing building systems, giving full functionality for smart building management. Supported protocols include: BACnet/IP, BACnet SC, Modbus/IP, Modbus RTU, P2, File data, Niagara, KNX, KNX PL-Link, SNMP, OPC UA, LoRaWAN, Modbus, M-Bus, LON, LBP, Diematic, Simatic S7 and IEC.

Once data is onboarded into the Building X platform, users can sort and clean it and then make it available to a wide variety of users and applications.[31]

3. How much maintenance **can be automated?**

A comprehensive ability to view buildings systems in the virtual world – both single structures and portfolios – triggers maintenance identification and fault or failure alerts to cloud dashboards. This is already an efficiency level improvement on most current O&M infrastructures. However, remote control takes efficiency a stage further again.

So critical assessment of potential solutions should not only examine interoperability and open architecture capabilities, but also real-life examples of remote issue resolution and AI-driven predictive maintenance. These will offer examples of actual savings rates achieved and will provide a credible basis on which to build your own business case.

Southern Methodist University (SMU) in Dallas, Texas, achieved 26% improvement in maintenance staff productivity by shifting from reactive to predictive maintenance campus-wide using Siemens Building X. This forward-looking approach helps enable precise prioritization of maintenance tasks, avoids breakdowns of key assets, and helps extend equipment life.

"There are over 12,000 measuring components out there across the campus that tell you there's going to be a problem or there is a problem in real time. You can remotely solve those problems without rolling a truck," explains Michael Molina - Vice President, Office of Facilities Planning & Management - University Architect, SMU

4. Can you create dashboards and visualizations that **fit your organization?**

Every building set-up is different, and solutions have to be able to offer a level of customization that truly fits the estate or portfolio in question. Stakeholders requiring dashboard visibility range from active estate management professionals, through occupier experience specifiers, to the CFO's office, each with their own particular requirements.

Specifically for the O&M team, there will be a variety of views needed. And those perspectives often need to be interrogated through cross-system data visualization tools. So it's important to know that you can track real-time KPIs, examine multi-layered views of assets/equipment for precise tracking, optimize maintenance schedules with historical trends and real-time analytics, streamline pre-maintenance planning with virtual walk-thoughts, and get actionable insights through advanced analytics to optimize uptime, asset utilization and equipment lifespan.

Analysis from the international property technology industry estimates that dirty data is typically costing over $800,000 a year for a 200,000 square foot commercial building.[32]

5. Can the **platform be scaled?**

For growing companies and portfolio managers, an autonomous platform needs to able to scale with the organization. Open architecture once more becomes a critical factor. No more vendor lock-in from different building system suppliers. And as new equipment assets are brought into the mix, the platform incorporates them into the unified view without disruption. This mitigates the risk of wholesale system replacement in the future.

Assessment of platforms should therefore evaluate whether a solution can scale robustly from one site to dozens or even hundreds. Facility teams are already stretched as a result of skills shortages and the increasing complexity of buildings technology. Scalable systems support is critical to trigger, co-ordinate, inform and optimize the deployment of scarce of human skills.

6. Is corporate sustainability and **energy efficiency reporting built-in?**

Corporate sustainability reporting is mandatory for larger companies in Europe and China. In the United States, the focus is on energy-efficiency. Whatever the label, all building owners and managers are interested in reducing energy costs.

In fact, the general point is that underperforming assets presenting faults and downtime are not energy efficient. So O&M need reporting to monitor and enhance building performance; and board directors need the same data to show the organization's track record on energy-efficiency and/or sustainability for external reporting and compliance.

However, the burden of capturing energy-efficiency and/or sustainability data and reporting it is considerable. It is a prime target for the benefits of digital data capture and automated report generation.

Report templates should be available to meet current standard formats – whether for progress tracking at board level, or at the level of detail required by regulators and financial markets.

The same is true for other compliance areas such as fire safety, helping O&M teams to concentrate on operational efficiency initiatives and not have their skills unnecessarily diverted with laborious administrative tasks.

The largest real estate investment company in Greece – PRODEA Investments - with a portfolio of 345 assets collaborates with Siemens Building X to fast track their sustainability goals. Their forward-thinking strategy is key in boosting tenant retention and increasing operational efficiency across their diverse portfolio. This digital platform enables them to effectively monitor and optimize energy consumption, costs, and emissions, while ensuring compliance with the latest EU environmental regulations.

Take Action

Are you ready to start your journey to O&M benefits and savings and discuss further to:

- Analyze how digitized your building or portfolio is
- Evaluate data interoperability
- Model savings & benefits
- Assess available autonomous smart solutions
- Create a continuous improvement strategy

Contact Us to discuss your transition →

Visit our website:

Explore how our Building X O&M offerings and real-world use cases can align seamlessly with your strategic business objectives. →

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